

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0162

Boeing Renton 2 100 kV
November 17, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0162. The Interconnection Customer proposes a 407.6 MW natural gas combined cycle facility interconnecting at the Boeing Renton 2 100 kV substation in the Indiana area. The request was received on October 05, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 122 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,289,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Collingwood – Allings Crossing 100.0 kV Line	Boeing Renton 2 – S Whitehall 100.0 kV	295.6	116	100.0
Collingwood – Boeing Renton 2 100.0 kV	South Whitehall Substation – Boeing Renton 2 100.0 kV	287.9	129	100.0
Boeing Renton 2 – South Whitehall 100.0 kV Line	Collingwood Substation – Boeing Renton 2 100.0 kV	262.6	141	100.0
Allings Crossing – Laotto 100.0 kV	Boeing Renton 2 – S Whitehall 100.0 kV	135.6	118	50.4
Swan Valley 500-230 kV Auto	Swan Valley – Trego 500.0 kV	102.7	881	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Collingwood – Allings Crossing 100.0 kV	Rebuild line with higher-capacity conductor	\$5,072,000	100% (\$5,072,000)
Collingwood – Boeing Renton 2 100.0 kV	Rebuild line with higher-capacity conductor	\$2,935,000	100% (\$2,935,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Whitehall – Boeing Renton 2 100.0 kV line	Rebuild line with higher-capacity conductor	\$5,835,000	100% (\$5,835,000)
Allings Crossing – Laotto 100.0 kV	Rebuild line with higher-capacity conductor	\$4,062,000	100% (\$4,062,000)
Swan Vlley 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$32,518,000	10% (\$3,385,000)
	Total allocated to project		\$21,289,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.